

September 17th, 2026

Curriculum Vitae

Shonosuke Takeshita

Theoretical Particle and Hadron Physics Group,
Physics Program, Graduate School of Advanced Science and Engineering,
Hiroshima University, Hiroshima 739-8526, Japan

PERSONAL DATA

First Name : Shonosuke
Last Name : Takeshita
E-mail : shonosuke@hiroshima-u.ac.jp
Nationality : Japan
Sex : male
Birth : Ehime, Japan, March 31st, 2000
Present Address: 1-3-1 Kagamiyama, Higashi-Hiroshima, Hiroshima 739-8526, Japan

ACADEMIC INTERESTS

Particle phenomenology and Cosmology:
Physics beyond the Standard Model, Supersymmetry, Axion physics, Leptogenesis, Grand Unified Theory

EDUCATION

April. 2024 - present : Doctor of science in Theoretical Physics
Hiroshima University, Japan

March. 2022 : Master of science in Theoretical Physics
Hiroshima University, Japan

March. 2022 : Bachelor of Science in Theoretical Physics
Hiroshima University, Japan

RESEARCH GRANTS

Apr. 2024 - Mar. 2027 : Program for Developing and Supporting the Next-Generation of Innovative Researchers
at Hiroshima University (SPRING)

AWARDS

Dec. 2025: Hiroshima University Excellent Student Scholarship,
Graduate School of Advanced Science and Engineering, Hiroshima University

Oct. 2025: Student Presentation Award of the Physical Society of Japan

Jun. 2025: Best Poster Award in “Higgs as a Probe of New Physics 2025 (HPNP2025)”
Selected by the poster jury (Chair: Prof. Michael Peskin)

Mar. 2025: HU SPRING Award 2024

Mar. 2025: Poster Award in “Workshop on Low BG Technologies 2024,”

Oct. 2024: Excellent poster in “HU SPRING 3QUESTIONS Three Questions for the Future”

Oct. 2023: Academic Encouragement Award, Graduate School of Advanced Science and Engineering,
Hiroshima University

Apr. 2023: Scholarships for Doctoral Programs, Faculty of Science, Hiroshima University

Dec. 2021: Encouragement Award from Support Group, Faculty of Science, Hiroshima University

TEACHING EXPERIENCE

Apr. 2022 - present : A School Committee Assistant, Information Media Center, Hiroshima University

Apr. 2023 - Aug. 2023 : Qualified Teaching Assistant, Hiroshima University

Apr. 2022 - Aug. 2022 : Qualified Teaching Assistant, Hiroshima University

PROFESSIONAL ACTIVITIES

Nov. 2022 - present : Member, The Physical Society of Japan

Publication List

Shonosuke Takeshita

Theoretical Particle and Hadron Physics Group,
Physics Program, Graduate School of Advanced Science and Engineering,
Hiroshima University, Hiroshima 739-8526, Japan

PUBLICATIONS

Under-Reviewed Manuscripts

1. **Shonosuke Takeshita** and Naoki Yoshioka, “Axionic Wormholes in Metric-Affine Gravity,” [\[arXiv:2609.11010 \[hep-ph\]\]](#).
2. Ryosuke Sato and **Shonosuke Takeshita**, “QCD axion from chiral gauge theories,” [\[arXiv:2508.21820 \[hep-ph\]\]](#).
3. Yusuke Shimizu and **Shonosuke Takeshita**, “Mass relations, Unification, and Proton Decay in the Mirror GUT Model,” [\[arXiv:2212.13064 \[hep-ph\]\]](#).

Peer-Reviewed Manuscripts

1. Yusuke Shimizu and **Shonosuke Takeshita**, “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,” *Nucl. Phys. B* **994** (2023), 116290, [\[arXiv:2303.11070 \[hep-ph\]\]](#).

PRESENTATIONS

International Conference

1. “QCD axion from chiral gauge theories,”
The 7th International Workshop on “Higgs as a Probe of New Physics 2025 (HPNP2025)” (Jun. 9th-13th, 2025), The University of Osaka, Osaka, Japan,
Jun. 10th, 2025, poster, **Best Poster Award**
2. “QCD axion from chiral gauge theories,”
KEK Theory Meeting on Particle Physics Phenomenology (KEK-PH2025winter) (Feb. 18th-21st, 2025), KEK, Tsukuba, Japan,
Feb. 21st, 2025, oral
3. “W boson mass and grand unification via the type-II seesaw-like mechanism,”
KEK Theory Meeting on Particle Physics Phenomenology (KEK-PH2023) (Nov. 7th-10th, 2023), KEK, Tsukuba, Japan,
Nov. 9th, 2023, oral
4. “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,”
27th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology (SI2023) (Aug. 21st-25th, 2023), SUN-LINK-SEA, Nantou, Taiwan,
Aug. 23rd, 2023, poster
5. “The W boson mass anomaly in the extension of the minimal SU(5) GUT model,”
3rd Hiroshima-IIB workshop in HEP (Feb. 20th-22nd, 2023), Hiroshima University, Hiroshima, Japan,
Feb. 21st, 2023, oral
6. “Experimental testability of GUT model including the mirror particles,”
KEK Theory Meeting on Particle Physics Phenomenology (KEK-PH2022) and International Joint Workshop on the Standard Model and beyond (Nov. 29th-Dec. 2nd, 2022), KEK, Tsukuba, Japan,
Dec. 1st, 2022, oral

National Conference

1. “QCD axion from chiral gauge theories and its phenomenology,”
JPS 2025 Annual (80th) meeting (Sep. 16th-19th, 2025), Hiroshima University, Hiroshima, Japan,
Sep. 16th, 2025, oral, **Student presentation Award**
2. “QCD axion from chiral gauge theories and its phenomenology,”
Progress in Particle Physics 2025 (PPP2025) (Sep. 1st-5th, 2025), YITP, Kyoto, Japan,
Sep. 1st, 2025, oral
3. “QCD axion from chiral gauge theories and its phenomenology,”
Summer school for Future Higgs Factories 2025 (Aug. 29th-31st, 2025), The University of Tokyo
Yamanaka-Ryo Naito Seminar House, Yamanashi, Japan,
Aug. 29th, 2025, oral
4. “QCD axion from chiral gauge theories and its phenomenology,”
10th Workshop on Low BG Technologies (Mar. 7th-8th, 2025), University of Toyama, Toyama,
Japan,
Mar. 7th, 2025, poster, **Poster Award**
5. “Solution to U(1) quality problem in Metric-Affine gravity,”
YONUPA Summer School 2024 (Aug. 21st-25th, 2024), Tokyo, Japan,
Aug. 23rd, 2024, poster
6. “W boson mass and grand unification via the type-II seesaw-like mechanism,”
New Higgs Working Group 37th Meeting (Jan. 20th, 2024), Osaka University, Osaka, Japan,
Jan. 20th, 2024, oral
7. “W boson mass and grand unification via the type-II seesaw-like mechanism,”
KEK-NAOJ Student Workshop 2023 (Dec. 16th-17th, 2023), online,
Dec. 16th, 2023, oral
8. “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,”
Flavor Physics workshop 2023 (FPWS2023) (Nov. 22nd-25th, 2023), Atami, Shizuoka, Japan,
Nov. 23rd, 2023, oral
9. “The prediction of the axion mass in the SU(5) GUT model,”
Setouchi Summer Institute 2023 (SSI2023) (Sep. 20st-22nd, 2023), Hiroshima University, Hi-
roshima, Japan,
Sep. 21st, 2023, oral
10. “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,”
JPS 2023 Annual (78th) meeting (Sep. 16th-19th, 2023), Tohoku University, Miyagi, Japan,
Sep. 17th, 2023, oral
11. “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,”
Progress in Particle Physics 2023 (PPP2023) (Aug. 28th-Sep. 1st, 2023), YITP, Kyoto, Japan,
Aug. 29th, 2023, poster
12. “W Boson Mass and Grand Unification via the Type-II Seesaw-like Mechanism,”
YONUPA Summer School 2023 (Aug. 17th-21st, 2023), Tokyo, Japan,
Aug. 20nd, 2023, oral
13. “The GUT model including the mirror particles and its experimental testability,”
JPS 2023 Spring (78th) meeting (Mar. 22nd-25th, 2023), Japan, online,
Mar. 23rd, 2023, oral
14. “The GUT model including the mirror particles and its experimental testability,”
Flavor Physics workshop 2022 (FPWS2022) (Nov. 7th-10th, 2022), Izunokuni City, Shizuoka,
Japan,
Nov. 9th, 2022, oral

15. “Experimental testability of GUT model including the mirror particles,”
Setouchi Summer Institute 2022 (SSI2022) (Sep. 26th-28th, 2022), Hiroshima University, Hiroshima, Japan,
Sep. 28th, 2022, oral
16. “The GUT model with observable proton decay,”
YONUPA Summer School 2022 (Aug. 6th-9th, 2022), Japan, [Review talk] (online),
Aug. 7th, 2022, oral

Seminar

1. “QCD axion from chiral gauge theories,”
Invited Talk, Nagoya University E-lab, Dec. 16th, 2025, oral
2. “QCD axion from chiral gauge theories and its phenomenology,”
Tokyo Metropolitan University, Mar. 14th, 2025, oral
3. “Axion quality problem in Metric-Affine gravity,”
Osaka University, Dec. 16th, 2024, oral